

Hanqing Zhu

Member of technical staff ◇ Science of RL ◇ xAI

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SUMMARY

I develop efficient and scalable learning algorithms for foundation models, with a focus on making capability acquisition more effective through methods that are both theoretically grounded and practically useful at scale.

I work at xAI on RL foundations in the **RL Science Team & Bigrun Team**. I am one of the key contributors to the latest Grok model, **Grok-4.5** for its RL scaling and core algorithmic changes, one of our biggest leaps.

Recently, I conducted the first study of RLVR training dynamics, aiming for a more interpretable view of the process and better optimization for improving intelligence. I also led the design of a highly memory-efficient optimizer for LLM training, recognized with an **Outstanding Paper Honorable Mention at MLSys 2025**.

EDUCATION

The University of Texas at Austin, Austin, TX, USA Aug. 2020 – May. 2026
Ph.D. in Electrical and Computer Engineering | First year (2020-2021) part-time in China
Advisor: [Prof. David Z. Pan](#)(ACM, IEEE, SPIE Fellow)
Thesis: *Co-Architecting Scalable Intelligence: From Hardware Substrates to Algorithms*

Shanghai Jiao Tong University, Shanghai, China Sept. 2016 – Jun. 2020
B.E. in Microelectronics Science and Engineering | Rank: 2nd/57
Graduated with Highest Honors

PROFESSIONAL EXPERIENCE

xAI, CA, USA Jan. 2026 – Present
Member of Technical Staff, RL Science & Coding RL Bigrun Team
▷ Core algorithms, training recipes, and optimization for Grok’s RL stack
▷ Key contributor to Grok 4.5 and Grok Build

Meta AI, CA, USA May 2025 – Dec 2025
Research Scientist Intern & Student Researcher, *Theory-driven Efficient and Scalable Learning for RLVR*
Mentor: [Dr. Yuandong Tian](#) (FAIR), [Dr. Zechun Liu](#)(Core AI), [Dr. Kai Sheng Tai](#) (Core AI)
▷ Led the first systematic study of RLVR training dynamics [M8]; follow-ups on LoRA initialization [M9] and RL-specific optimization [M10] (data-free on-policy distillation and stronger RL optimizers)

Meta AI, CA, USA May 2024 – Oct 2024
Research Scientist Intern, *Efficient and Scalable Large-scale Pre-Training*
Mentor: [Dr. Jinwon Lee](#)
▷ The State-of-the-art Memory-efficient training for large language models [M5] (**MLSys’25 Outstanding Paper Honorable Mention**); integrated into HuggingFace, LLaMA-Factory, Axolotl, and FluxML.

Lightelligence Inc., MA, USA May 2023 – Sept 2023
Software Research Intern, *Low-bit Chip-aware Training*
Mentor: [Dr. Weifeng Zhang](#)
▷ Low-precision noise-aware training for state-of-the-art photonic AI accelerators.

Google Brain (now Google Deepmind), CA, USA Jul 2022 – Nov 2022
Student Researcher, *RL-based Chip Placement*
Mentor: [Dr. Joe Jiang](#)
▷ Chip placement with reinforcement learning ([AlphaChip](#)). Integrate [DREAMPlace](#) for the RL chip placer.

AWARDS AND HONORS

Best Paper Award [M7]	CVPR AI4CC Workshop	2025
ICLR notable reviewer	ICLR	2025
Outstanding Paper Honorable Mention [M5]	MLSys	2025
DAC Ph.D. Forum	DAC	2025
MLSys Student Travel Award	MLSys	2025
ML and Systems Rising Stars (38 worldwide)	MLCommons	2025
Texas ECE Graduate Achievement Award	UT Austin	2024
UT Graduate School Continuing Fellowship Nomination	UT Austin	2024

(1 of 2 nominees in the entire ECE department)

1st Place in IEEE/ACM MLCAD FPGA Macro-Placement Contest	MLCAD	2023
MLSys Student Travel Award	MLSys	2023
Winner of Robert S. Hilbert Memorial Optical Design Competition	Synopsys	2022
DAC Young Fellow	DAC	2021
Shanghai Outstanding Graduate	Shanghai City	2020
Departmental Excellent Undergraduate Thesis	SJTU	2020
Hongyi Scholarship	SJTU	2019
Outstanding Undergraduate Scholarship	SJTU	2019
Samsung Scholarship	SJTU	2018
Zhiyuan College Honors Scholarship	SJTU	2018
1st Prize, National Mathematical Contest in Modeling	Shanghai Division	2018
Academic Excellence Scholarship	SJTU	2017-2019

PROFESSIONAL SERVICE

- ▷ **Conference Reviewer:** ICML, NeurIPS, ICLR, CVPR, AAAI, DAC, ICCAD, FPGA, AICAS
- ▷ **Journal Reviewer:** Nature Communications, TNNLS, TCAD, Journal of Applied Physics, Photonic Network Communications

FEATURED PUBLICATIONS

I have published papers in top conferences in machine learning/ system/computer architecture/design automation, including MLSys, HPCA, NeurIPS, ICCV, COLM, DAC, ICCAD, and TCAD.

Representative publications that I am a primary author of. (* denoted co-first author ; ♠ denote equal advisory)

Efficient, Scalable and Hardware-aware AI Algorithms

- [M10] **Hanqing Zhu**, Wenyan Cong, Zhizhou Sha, Sagnik Mukherjee, Xinyuan Song, Xiaoxia Wu, Yuandong Tian, Shiwei Liu, David Z. Pan, Zhangyang Wang “ ISO: Isospectral Optimization for RLVR ” In submission. [[First RL optimizer](#)]
- [M9] Ruijia Zhang, Jiacheng Zhu, **Hanqing Zhu**, Laixi Shi “ Geometry-Preserving Orthonormal Initialization for Low-Rank Adaptation in Reinforcement Learning ” In *International Conference on Machine Learning (ICML)*, 2026 [[RL training dynamics-guided LoRA](#)]
- [M8] **Hanqing Zhu**, Zhenyu Zhang, Hanxian Huang, DiJia Su, Zechun Liu, Jiawei Zhao, Igor Fedorov, Hamed Pirsiavash, Jinwon Lee, David Z. Pan, Zhangyang Wang[♠], Yuandong Tian[♠], Kai Sheng Tai[♠], “ The Path Not Taken: RLVR Provably Learns Off the Principals ” In submission. Preliminary version accepted by NeurIPS 2025 Workshop on Efficient Reasoning (**Spotlight**). [[Paper](#); [X post](#); [量子位](#) ; [新智元](#) ; [First theory-driven RLVR study and guidance for geometry-aligned RL optimization](#)]
- [M7] Wenyan Cong, **Hanqing Zhu**, Kevin Wang, Jiahui Lei, Colton Stearns, Yuanhao Cai, Dilin Wang, Rakesh Ranjan, Matt Feiszli, Leonidas Guibas, Zhangyang Wang, Weiyao Wang, Zhiwen Fan “ VideoLifter: Lifting Videos to 3D with Fast Hierarchical Stereo Alignment ” in *International Conference on 3D Vision*, 2026; CVPR 2025 Workshop on AI for Content Creation (**Oral Presentation**; **♣Best Paper Award**) [[Paper](#); [Code](#); [Efficient video-3D reconstruction](#).]
- [M6] Wenyan Cong*, **Hanqing Zhu***, Peihao Wang, Bangya Liu, Dejia Xu, Kevin Wang, David Z. Pan, Yan Wang, Zhiwen Fan, Zhangyang Wang, “ Can Test-Time Scaling Improve World Foundation Model? ” In *Conference on Language Modeling (COLM)*, 2025. [[Paper](#); [Code](#); [First efficient test-time scaling for world foundation model](#)]
- [M5] **Hanqing Zhu***, Zhenyu Zhang*, Wenyan Cong, Xi Liu, Sem Park, Vikas Chandra, Bo Long, David Z. Pan, Zhangyang Wang, Jinwon Lee. “ APOLLO: SGD-like Memory, AdamW-level Performance. ” in *Conference on Machine Learning and Systems (MLSys)*, 2025 [**♣Outstanding Paper Honorable Mention**; [Paper](#); [Code](#), [250+ stars](#); [Hacker News](#); [Huggingface Transformers](#); [LLaMA-Factory](#); [FluxML](#); [axolotl](#); [机器之心](#) ; [Theory-driven scalable memory-efficient training with new-recording memory efficiency](#)]
- [M4] **Hanqing Zhu**, Wenyan Cong, Guojin Chen, Shupeng Ning, Ray Chen, Jiaqi Gu, and David Z. Pan, “ PACE: Pacing Operator Learning to Accurate Optical Field Simulation for Complicated Photonic Devices, ” in *Conference on Neural Information Processing Systems (NeurIPS)*, 2024 [[Paper](#); [Code](#); [Theory-grounded efficient and fast operator model for scientific simulation](#)]
- [M3] Zixuan Jiang, Jiaqi Gu, **Hanqing Zhu**, and David Z. Pan, “ Pre-RMSNorm and Pre-CRMSNorm Transformers: Equivalent and Efficient Pre-LN Transformers, ” in *Conference on Neural Information Processing Systems (NeurIPS)*, Dec 10 - Dec 16, 2023 (**Spotlight**). (Acceptance Rate: 26.1%) [[Paper](#); [code](#); [Theory-grounded efficient Pre-LN implementation](#)]
- [M2] Jiaqi Gu, **Hanqing Zhu**, Chenghao Feng, Zixuan Jiang, Ray T. Chen, and David Z. Pan, “ L2ight: Enabling On-Chip Learning for Optical Neural Networks via Efficient in-situ Subspace Optimization, ” in *Conference on Neural Information Processing Systems (NeurIPS)*, Dec., 2021 [[Paper](#); [Code](#); [Hardware-aware efficient on-chip training](#)]

- [M1] Jiaqi Gu, **Hanqing Zhu**, Chenghao Feng, Mingjie Liu, Zixuan Jiang, Ray T. Chen, and David Z. Pan, "Towards Memory-Efficient Neural Networks via Multi-Level in situ Generation," in *International Conference on Computer Vision (ICCV)*, Oct., 2021 [[Paper](#); [Low-rank decomposition for memory-efficiency](#)]

Hardware-software Co-design for Efficient AI Hardware/Systems

- [I11] **Hanqing Zhu**, Shupeng Ning, Hongjian Zhou, Ziang Yin, Ray T Chen, Jiaqi Gu, David Z Pan " Harnessing Photonics for Machine Intelligence " In *IEEE Circuits and Systems Magazine*, 2026, Accepted]
- [I10] **Hanqing Zhu**, Zhican Zhou, Shupeng Ning, Xuhao Wu, Ray Chen, Yating Wan and David Z. Pan, " ENLighten: Lighten the Transformer, Enable Efficient Optical Acceleration " In *IEEE/ACM Asia and South Pacific Design Automation Conference (ASP-DAC)*, 2026. [[Paper](#); [Low-rank plus hardware-aware sparsity decomposition](#)]
- [I9] **Hanqing Zhu**, Jiaqi Gu, Hanrui Wang, Zixuan Jiang, Zhekai Zhang, Rongxin Tang, Chenghao Feng, Song Han, Ray T. Chen, David Z. Pan, "Lightening-Transformer: A Dynamically-operated Optically-interconnected Photonic Transformer Accelerator," in *IEEE International Symposium on High Performance Computer Architecture(HPCA)*, Mar. 2024 (Acceptance Rate: 18.3% (75 of 410)) [[Paper](#); [Code](#); [Hardware-software Co-design](#)]
- [I8] **Hanqing Zhu**, Jiaqi Gu, Hanrui Wang, Rongxin Tang, Zhekai Zhang, Chenghao Feng, Song Han, Ray T. Chen, David Z. Pan, " DOTA: A Dynamically-Operated Photonic Tensor Core for Energy-Efficient Transformer Accelerator," in *Conference on Machine Learning and Systems (MLSys), Workshop on Systems for Next-Gen AI Paradigms (SNAP)*, Jun 4 - Jun 8, 2023
- [I7] **Hanqing Zhu**, Keren Zhu, Jiaqi Gu, Harrison Jin, Ray Chen, Jean Anne Incorvia and David Z. Pan, "Fuse and Mix: MACAM-Enabled Analog Activation for Energy-Efficient Neural Acceleration" in *IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, Oct. 2022 [[Paper](#); [NAS with mixed activation precision](#)]
- [I6] **Hanqing Zhu**, Jiaqi Gu, Chenghao Feng, Mingjie Liu, Zixuan Jiang, Ray T. Chen, and David Z. Pan, "ELight: Enabling Efficient Photonic In-Memory Neurocomputing with Life Enhancement," in *IEEE/ACM Asia and South Pacific Design Automation Conference (ASP-DAC)*, Jan., 2022 [[Paper](#); [Hardware-software Co-design](#)]
- [I5] Jiaqi Gu, **Hanqing Zhu**, Chenghao Feng, Zixuan Jiang, Mingjie Liu, Shuhan Zhang, Ray T. Chen, and David Z. Pan, "ADEPT: Automatic Differentiable DEsign of Photonic Tensor Cores," in *ACM/IEEE Design Automation Conference (DAC)*, Jul., 2022 [[Paper](#); [NAS for automatic efficient hardware search](#)]
- [I4] **Hanqing Zhu***, Shupeng Ning*, Chenghao Feng, Jiaqi Gu, Zhixing Jiang, Zhoufeng Ying, Jason Midkiff, Sourabh Jain, May H. Hlaing, David Z. Pan, and Ray T. Chen, "Photonic-Electronic Integrated Circuits for High-Performance Computing and AI Accelerator," in *IEEE Journal of Lightwave Technology (JLT)*, July. 2024 [[Paper](#);]
- [I3] Jiaqi Gu, **Hanqing Zhu**, Chenghao Feng, Zixuan Jiang, Ray T. Chen, and David Z. Pan, "M3ICRO: Machine Learning-Enabled Compact Photonic Tensor Core based on PROgrammable Multi-Operand Multimode Interference," in *APL Machine Learning*, Jan. 2024
- [I2] Harrison Jin, **Hanqing Zhu**, Keren Zhu, Thomas Leonard, Jaesuk Kwon, Mahshid Alamdar, Kwangseok Kim, Jungsik Park, Naoki Hase, David Z. Pan, Jean Anne C. Incorvia, "Domain Wall-Magnetic Tunnel Junction Analog Content Addressable Memory Using Current and Projected Data" in *IEEE Transactions on Nanotechnology*, 2024
- [I1] **Hanqing Zhu**, Jiaqi Gu, Chenghao Feng, Mingjie Liu, Zixuan Jiang, Ray T. Chen, and David Z. Pan, "ELight: Towards Efficient and Aging-Resilient Photonic In-Memory Neurocomputing," in *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, Jun., 2022 [[Paper](#);]